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Lab Report 3:

**Study of Linear and Circular Convolution, Cross-Correlation, and Auto-Correlation
of Discrete Signals**

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Study of Linear and Circular Convolution, Cross-Correlation, and Auto-Correlation of Discrete Signals

Theory

In digital signal processing, convolution and correlation are fundamental operations for analyzing and manipulating discrete-time signals [1, 2, 3]. These techniques are widely used in system analysis, filtering, and pattern recognition.

1. Linear Convolution:

Linear convolution combines two discrete signals to produce a third signal that represents the amount of overlap between the signals as one slides past the other. For signals $x[n]$ and $h[n]$, the linear convolution is defined as:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n - k] \quad (1)$$

Linear convolution is used to determine the output of linear time-invariant (LTI) systems and is essential in filtering applications.

2. Circular Convolution:

Circular convolution assumes the signals are periodic and wraps around the ends. For two sequences of length N , the circular convolution is:

$$y[n] = \sum_{k=0}^{N-1} x[k] \cdot h[(n - k) \bmod N] \quad (2)$$

Circular convolution is important in fast algorithms such as the Fast Fourier Transform (FFT) and in systems where signals are inherently periodic.

3. Cross Correlation:

Cross correlation measures the similarity between two signals as one is shifted over the other. For $x[n]$ and $y[n]$:

$$r_{xy}[m] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n + m] \quad (3)$$

It is used in pattern matching, signal detection, and time delay estimation.

4. Auto Correlation:

Auto correlation is the cross correlation of a signal with itself. For $x[n]$:

$$r_{xx}[m] = \sum_{n=-\infty}^{\infty} x[n] \cdot x[n + m] \quad (4)$$

Auto correlation is used to analyze signal properties such as periodicity and to detect repeating patterns.

Plotting and Comparison:

By plotting the results of linear convolution, circular convolution, cross correlation, and auto correlation for a given signal $x(n)$, differences in their behavior can be observed. Linear convolution typically produces a longer output sequence than circular convolution, which wraps around and maintains the original length. Correlation operations reveal similarities and periodicities in signals, with auto correlation highlighting self-similarity and cross correlation showing the relationship between two signals. Understanding these differences is crucial for selecting the appropriate operation in practical signal processing tasks.

Used Tools

- Matlab Online
- LaTeX

Code & Output

1. Convolution and Correlation of Discrete Signals

```
1  % Given signals
2  x = [2, 1, 0, -1, 2, 5, 6];
3  h = [0, 0, 2, 2, 4, 5, 6];
4
5  % Linear convolution
6  lin_conv = conv(x, h);
7
8  % Circular convolution (length =
9  %   max(length(x), length(h)))
10 N = max(length(x), length(h));
11 circ_conv = cconv(x, h, N);
12
13 % Cross-correlation
14 cross_corr = xcorr(x, h);
15
16 % Auto-correlation of x(n)
17 auto_corr = xcorr(x);
18
19 % Plotting
20 figure;
21 subplot(3,2,1);
22 stem(0:length(x)-1, x, 'filled');
23 title('Original Signal x(n)');
24 xlabel('n'); ylabel('Amplitude');
25
26 subplot(3,2,2);
27 stem(0:length(h)-1, h, 'filled');
28 title('Original Signal h(n)');
29 xlabel('n'); ylabel('Amplitude');
30
31 subplot(3,2,3);
32 stem(0:length(lin_conv)-1, lin_conv,
33     'filled');
34 title('Linear Convolution');
35 xlabel('n'); ylabel('Amplitude');
36
37 subplot(3,2,4);
38 stem(0:length(circ_conv)-1, circ_conv,
39     'filled');
40 title('Circular Convolution');
41 xlabel('n'); ylabel('Amplitude');
42
43 subplot(3,2,5);
44 stem(-(length(x)-1):(length(h)-1),
45     cross_corr, 'filled');
46 title('Cross Correlation');
47 xlabel('Lag'); ylabel('Amplitude');
48
49 subplot(3,2,6);
50 stem(-(length(x)-1):(length(x)-1),
51     auto_corr, 'filled');
52 title('Auto Correlation of x(n)');
53 xlabel('Lag'); ylabel('Amplitude');
54
55 sgtitle('Signal Operations: Original,
56     Convolution and Correlation');
```

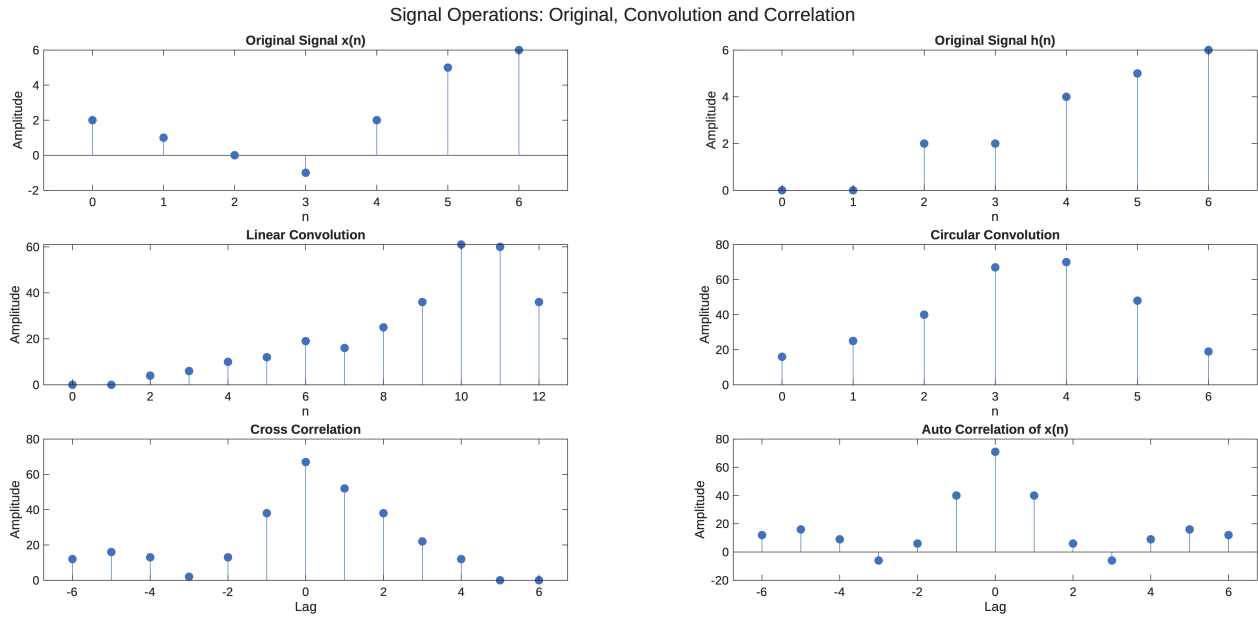


Figure 1: Operations on $x(n)$ and $h(n)$

Comments on Differences

- Linear convolution gives a longer output (length = $L + M - 1$) and represents the true convolution in the time domain.
- Circular convolution assumes periodicity and wraps around, so the result may differ from linear convolution unless signals are zero-padded.
- Cross-correlation measures similarity between $x(n)$ and $h(n)$ as one is shifted over the other.
- Auto-correlation measures similarity of $x(n)$ with itself, peaking at zero lag.

2. Convolution and Correlation of Discrete Sinusoidal Signals

```

1  % Parameters
2  cycles = 2;                % Number of
   ↳ cycles
3  samples_per_cycle = 50;    % Increased
   ↳ sample points per cycle for smoothness
4  N = cycles * samples_per_cycle;
5  n = 0:N-1;
6
7  % Define signals
8  theta = 2*pi*n/samples_per_cycle;
9  x = sin(theta);
10 h = cos(theta);
11
12 % Linear Convolution
13 lin_conv = conv(x, h);
14
15 % Circular Convolution (length = N)
16 circ_conv = cconv(x, h, N);
17
18 % Cross Correlation
19 cross_corr = xcorr(x, h);
20
21 % Auto Correlation of x(n)
22 auto_corr = xcorr(x);
23
24 % Plotting
25 figure;
26
27 subplot(3,2,1);
28 stem(n, x, 'filled');
29 title('Original Signal x(n) = sin(n)');
30 xlabel('n'); ylabel('Amplitude');
31
32 subplot(3,2,2);
33 stem(n, h, 'filled');
34 title('Original Signal h(n) = cos(n)');
35 xlabel('n'); ylabel('Amplitude');
36
37 subplot(3,2,3);

```

```

38 stem(0:length(lin_conv)-1, lin_conv,
    ⇐ 'filled');
39 title('Linear Convolution');
40 xlabel('n'); ylabel('Amplitude');
41
42 subplot(3,2,4);
43 stem(0:length(circ_conv)-1, circ_conv,
    ⇐ 'filled');
44 title('Circular Convolution');
45 xlabel('n'); ylabel('Amplitude');
46
47 subplot(3,2,5);
48 stem(-(length(x)-1):(length(h)-1),
    ⇐ cross_corr, 'filled');
49 title('Cross Correlation');
50 xlabel('Lag'); ylabel('Amplitude');
51
52 subplot(3,2,6);
53 stem(-(length(x)-1):(length(x)-1),
    ⇐ auto_corr, 'filled');
54 title('Auto Correlation of x(n)');
55 xlabel('Lag'); ylabel('Amplitude');
56
57 sgtitle('Signal Operations: Sine and Cosine
    ⇐ Signals');

```

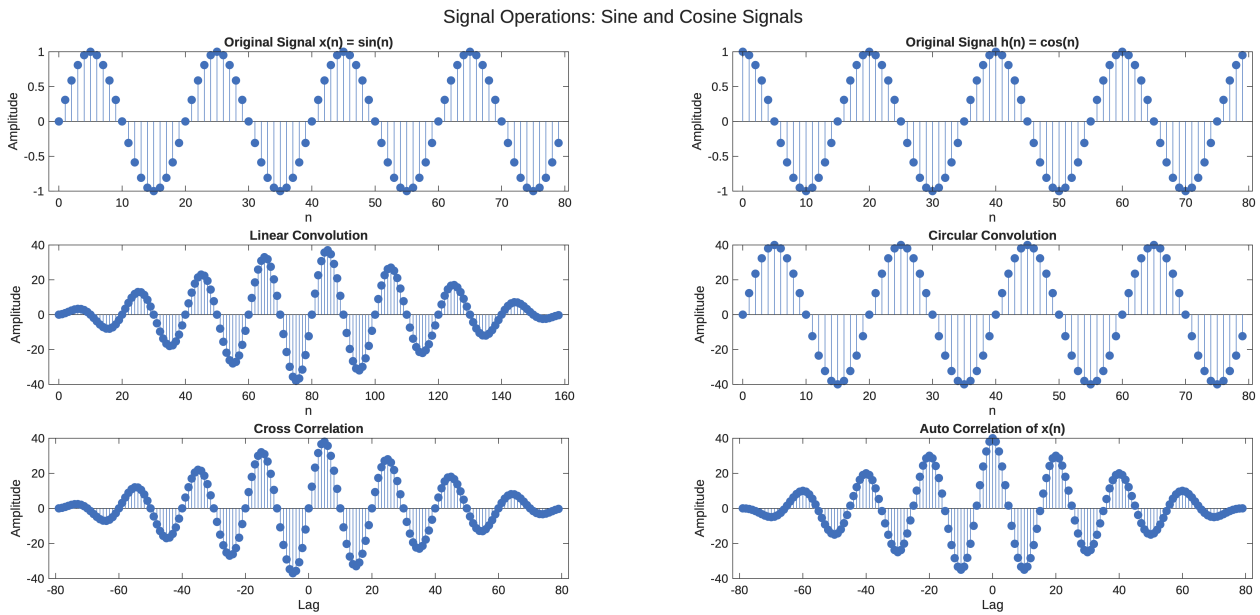


Figure 2: Operations on $x(n) = \sin(n)$ and $h(n) = \cos(n)$

Comments on Differences

- Linear convolution gives the true time-domain result; the output is longer than the input signals.
- Circular convolution wraps around and assumes periodicity, so the result may differ from linear convolution.
- Cross-correlation shows similarity between the two signals as one slides over the other.
- Auto-correlation peaks at zero lag, indicating self-similarity and periodicity.

3. Convolution and Correlation of Transformed Discrete Signals I

```

1 % Given signals
2 x = [2, 1, 0, -1, 2, 5, 6];
3 h = [0, 0, 2, 2, 4, 5, 6];
4
5 % Operation 1: x(3n-2)
6 n = 1:length(x);
7 n1 = 1:ceil((length(x)+2)/3); % Find valid
    ⇐ n for x(3n-2)
8 x_3n_2 = zeros(1, length(n1));
9 for i = 1:length(n1)

```

```

10     idx = 3*n1(i)-2;
11     if idx >= 1 && idx <= length(x)
12         x_3n_2(i) = x(idx);
13     end
14 end
15
16 % Operation 2: -h(n+2)
17 n2 = 1:length(h);
18 h_n_plus_2 = zeros(1, length(h));
19 for i = 1:length(h)
20     idx = i+2;
21     if idx <= length(h)
22         h_n_plus_2(i) = -h(idx);
23     else
24         h_n_plus_2(i) = 0;
25     end
26 end
27
28 % Linear convolution
29 lin_conv = conv(x_3n_2, h_n_plus_2);
30
31 % Circular convolution (length = max of
32   ↳ input lengths)
33 N = max(length(x_3n_2),
34   ↳ length(h_n_plus_2));
35 circ_conv = cconv(x_3n_2, h_n_plus_2, N);
36
37 % Cross correlation
38 cross_corr = xcorr(x_3n_2, h_n_plus_2);
39
40 % Auto correlation on x(n)
41 auto_corr = xcorr(x_3n_2);
42
43 % Plotting
44 figure;
45 subplot(3,2,1);
46 stem(x_3n_2, 'filled');
47 title('x(3n-2)');
48 xlabel('n'); ylabel('Amplitude');
49
50 subplot(3,2,2);
51 stem(h_n_plus_2, 'filled');
52 title('-h(n+2)');
53 xlabel('n'); ylabel('Amplitude');
54
55 subplot(3,2,3);
56 stem(lin_conv, 'filled');
57 title('Linear Convolution');
58 xlabel('n'); ylabel('Amplitude');
59
60 subplot(3,2,4);
61 stem(circ_conv, 'filled');
62 title('Circular Convolution');
63 xlabel('n'); ylabel('Amplitude');
64
65 subplot(3,2,5);
66 stem(cross_corr, 'filled');
67 title('Cross Correlation');
68 xlabel('n'); ylabel('Amplitude');
69
70 subplot(3,2,6);
71 stem(auto_corr, 'filled');
72 title('Auto Correlation of x(3n-2)');
73 xlabel('n'); ylabel('Amplitude');
74
75 sgtitle('DSP Operations Results');

```

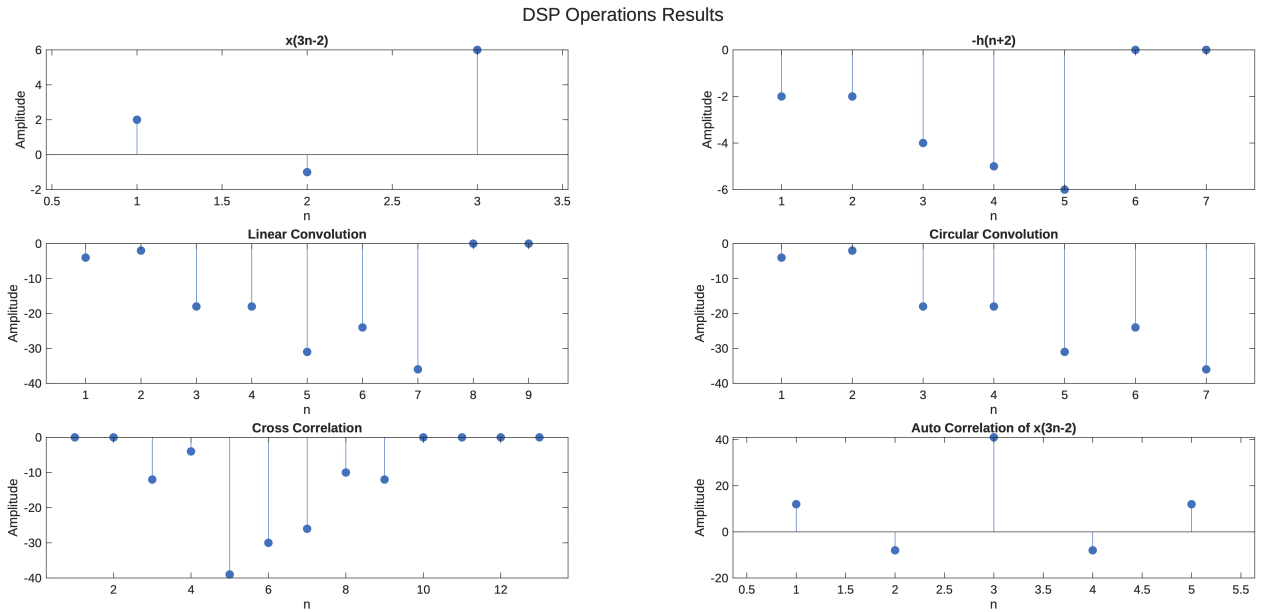


Figure 3: Operations on $x(3n - 2)$ and $-h(n + 2)$

Comments on Differences

- Linear convolution gives the full convolution result, including edge effects.
- Circular convolution wraps around, so results differ at the edges.
- Cross-correlation measures similarity between two signals as one is shifted over the other.
- Auto-correlation measures similarity of a signal with itself, peaking at zero lag.

4. Convolution and Correlation of Transformed Discrete Signals II

```
1  % Given signals
2  x = [2, 1, 0, -1, 2, 5, 6];
3  h = [0, 0, 2, 2, 4, 5, 6];
4
5  % Operation 1: x(-n+4)
6  N_x = length(x);
7  n_x = 0:N_x-1;
8  x_neg_n_plus_4 = zeros(1, N_x);
9  for i = 1:N_x
10     idx = -n_x(i) + 4 + 1; % +1 for MATLAB
11     % indexing
12     if idx >= 1 && idx <= N_x
13         x_neg_n_plus_4(i) = x(idx);
14     else
15         x_neg_n_plus_4(i) = 0;
16     end
17 end
18 % Operation 2: h(-n-3)
19 N_h = length(h);
20 n_h = 0:N_h-1;
21 h_neg_n_minus_3 = zeros(1, N_h);
22 for i = 1:N_h
23     idx = -n_h(i) - 3 + 1; % +1 for MATLAB
24     % indexing
25     if idx >= 1 && idx <= N_h
26         h_neg_n_minus_3(i) = h(idx);
27     else
28         h_neg_n_minus_3(i) = 0;
29     end
30 end
31 % Linear convolution
32 lin_conv = conv(x_neg_n_plus_4,
33     % h_neg_n_minus_3);
34 % Circular convolution (length = max of
35     % input lengths)
36 N = max(length(x_neg_n_plus_4),
37     % length(h_neg_n_minus_3));
38 circ_conv = cconv(x_neg_n_plus_4,
39     % h_neg_n_minus_3, N);
37
38 % Cross correlation
39 cross_corr = xcorr(x_neg_n_plus_4,
40     % h_neg_n_minus_3);
41
42 % Auto correlation on x(-n+4)
43 auto_corr = xcorr(x_neg_n_plus_4);
44
45 % Plotting
46 figure;
47 subplot(3,2,1);
48 stem(x_neg_n_plus_4, 'filled');
49 title('x(-n+4)');
50 xlabel('n'); ylabel('Amplitude');
51 subplot(3,2,2);
52 stem(h_neg_n_minus_3, 'filled');
53 title('h(-n-3)');
54 xlabel('n'); ylabel('Amplitude');
55 subplot(3,2,3);
56 stem(lin_conv, 'filled');
57 title('Linear Convolution');
58 xlabel('n'); ylabel('Amplitude');
59 subplot(3,2,4);
60 stem(circ_conv, 'filled');
61 title('Circular Convolution');
62 xlabel('n'); ylabel('Amplitude');
63 subplot(3,2,5);
64 stem(cross_corr, 'filled');
65 title('Cross Correlation');
66 xlabel('n'); ylabel('Amplitude');
67 subplot(3,2,6);
68 stem(auto_corr, 'filled');
69 title('Auto Correlation of x(-n+4)');
70 xlabel('n'); ylabel('Amplitude');
71
72 sgtitle('DSP Operations Results for x(-n+4)
73     % and h(-n-3)');
```

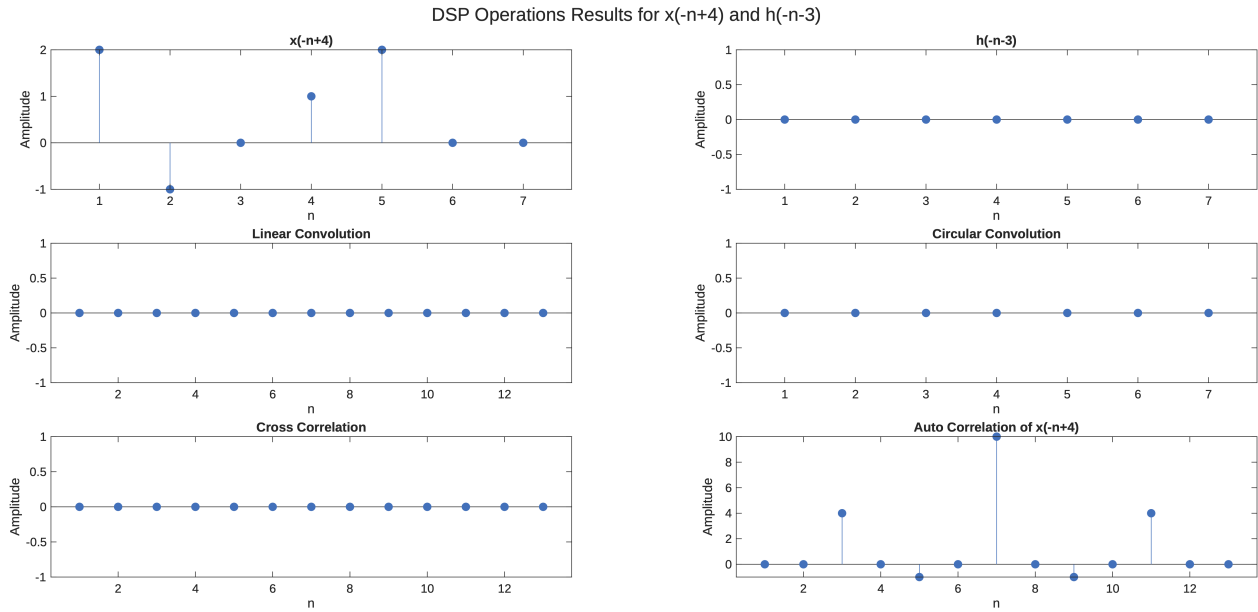


Figure 4: Operations on $x(-n+4)$ and $h(-n-3)$

Comments on Differences

- Linear convolution gives a longer, true result; circular wraps and may cause aliasing unless zero-padded.
- Measures similarity between $x(-n+4)$ and $h(-n-3)$ as one shifts over the other.
- Measures self-similarity of $x(-n+4)$, peaking at zero lag.

5. Convolution and Correlation of Transformed Discrete Sinusoidal Signals I

```

1  % Parameters
2  cycles = 4;                % Number of
   ↳ cycles
3  samples_per_cycle = 50;    % Increased
   ↳ sample points per cycle for smoothness
4  N = cycles * samples_per_cycle;
5  n = 0:N-1;
6
7  % Define signals
8  theta = 2*pi*n/samples_per_cycle;
9  x = sin(theta);
10 h = cos(theta);
11
12
13 % x(3n-2)
14 n1 = 1:ceil((N+2)/3);
15 x_3n_2_full = zeros(1, length(n1));
16 for i = 1:length(n1)
17     idx = 3*n1(i)-2;
18     if idx >= 1 && idx <= N
19         x_3n_2_full(i) = x(idx);
20     end
21 end
22
23 valid_idx = find(x_3n_2_full ~= 0 | n1 ==
   ↳ 1);
24 x_3n_2 = x_3n_2_full(valid_idx);
25 n1_valid = n1(valid_idx);
26
27 % -h(n+2)
28 h_n_plus_2 = zeros(1, length(x_3n_2));
29 for i = 1:length(x_3n_2)
30     idx = i+2;
31     if idx <= N
32         h_n_plus_2(i) = -h(idx);
33     else
34         % If out of bounds, pad with next
35         ↳ values from h (wrap around or
36         ↳ repeat last value)
37         pad_idx = mod(idx-1, N) + 1;
38         h_n_plus_2(i) = -h(pad_idx);
39     end
40 end
41
42 % Now x_3n_2 and h_n_plus_2 are the same
43 ↳ length for all operations

```

```

41 % Linear convolution
42 lin_conv = conv(x_3n_2, h_n_plus_2);
43
44 % Circular convolution (length = max of
45   ↪ input lengths)
46 circ_conv = cconv(x_3n_2, h_n_plus_2, Nc);
47
48 % Cross correlation
49 cross_corr = xcorr(x_3n_2, h_n_plus_2);
50
51 % Auto correlation on x(3n-2)
52 auto_corr = xcorr(x_3n_2);
53
54 % Plotting
55 figure;
56 subplot(3,2,1);
57 stem(n1_valid, x_3n_2, 'filled');
58 title('x(3n-2) [sin]');
59 xlabel('n'); ylabel('Amplitude');
60
61 subplot(3,2,2);
62 stem(1:length(h_n_plus_2), h_n_plus_2,
63   ↪ 'filled');
64 title('-h(n+2) [cos]');
65 xlabel('n'); ylabel('Amplitude');
66
67 subplot(3,2,3);
68 stem(0:length(lin_conv)-1, lin_conv,
69   ↪ 'filled');
70 title('Linear Convolution');
71 xlabel('n'); ylabel('Amplitude');
72
73 subplot(3,2,4);
74 stem(0:length(circ_conv)-1, circ_conv,
75   ↪ 'filled');
76 title('Circular Convolution');
77 xlabel('n'); ylabel('Amplitude');
78
79 subplot(3,2,5);
80 stem(-(length(x_3n_2)-1):(length(h_n_plus_2)-1),
81   ↪ cross_corr, 'filled');
82 title('Cross Correlation');
83 xlabel('Lag'); ylabel('Amplitude');
84
85 subplot(3,2,6);
86 stem(-(length(x_3n_2)-1):(length(x_3n_2)-1),
87   ↪ auto_corr, 'filled');
88 title('Auto Correlation of x(3n-2)');
89 xlabel('Lag'); ylabel('Amplitude');
90
91 sgtitle('Signal Operations: Transformed
92   ↪ Sine and Cosine');

```

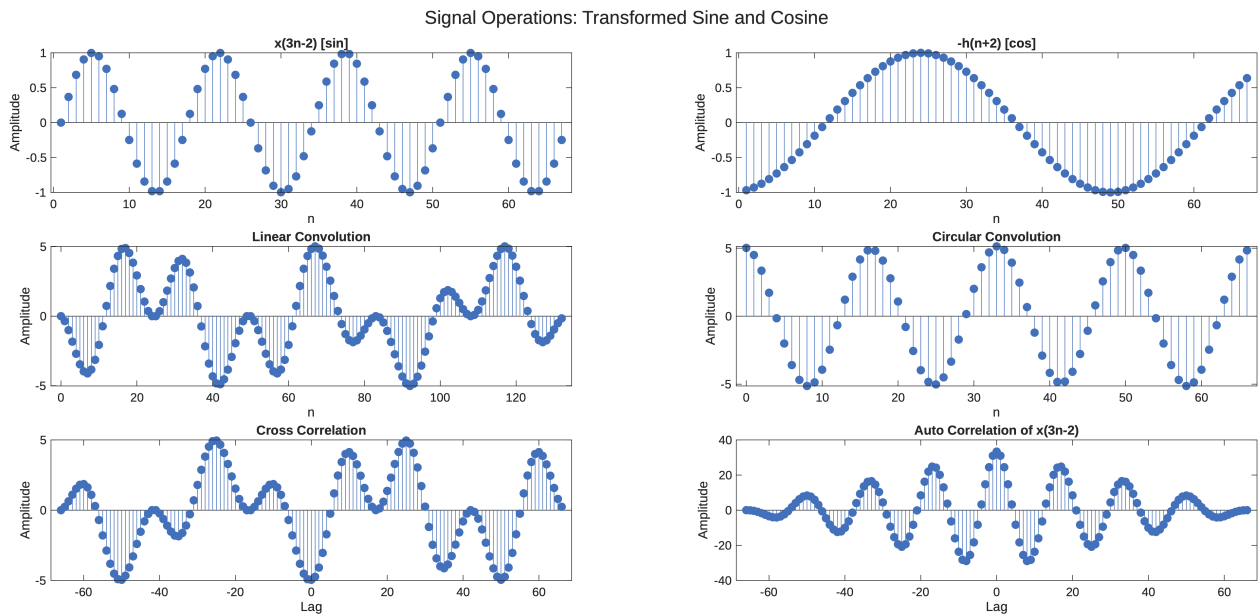


Figure 5: Operations on $x(3n-2)$ and $-h(n+2)$

Comments on Differences

- Linear convolution: true time-domain result, longer output.
- Circular convolution: wraps around, assumes periodicity.

- Cross-correlation: shows similarity as signals shift.
- Auto-correlation: peaks at zero lag, self-similarity.

6. Convolution and Correlation of Transformed Discrete Sinusoidal Signals II

```

1  % Parameters
2  cycles = 4;           % Number of
   ↪ cycles
3  samples_per_cycle = 50; % Increased
   ↪ sample points per cycle for smoothness
4  N = cycles * samples_per_cycle;
5  n = 0:N-1;
6
7  % Define signals
8  theta = 2*pi*n/samples_per_cycle;
9  x = sin(theta);
10 h = cos(theta);
11
12 % x(-n+4)
13 n2 = 0:N-1;
14 idx_x = -n2 + 4 + 1; % +1 for MATLAB
   ↪ indexing
15 x_shifted = zeros(1, N);
16 for i = 1:N
17     if idx_x(i) >= 1 && idx_x(i) <= N
18         x_shifted(i) = x(idx_x(i));
19     else
20         % Pad with wrapped values from x
21         pad_idx = mod(idx_x(i)-1, N) + 1;
22         x_shifted(i) = x(pad_idx);
23     end
24 end
25
26 % h(-n-3)
27 idx_h = -n2 - 3 + 1;
28 h_shifted = zeros(1, N);
29 for i = 1:N
30     if idx_h(i) >= 1 && idx_h(i) <= N
31         h_shifted(i) = h(idx_h(i));
32     else
33         pad_idx = mod(idx_h(i)-1, N) + 1;
34         h_shifted(i) = h(pad_idx);
35     end
36 end
37
38 % Now x_shifted and h_shifted are the same
   ↪ length for all operations
39
40 % Linear convolution
41 lin_conv = conv(x_shifted, h_shifted);
42
43 % Circular convolution (length = N)
44 Nc = N;
45 circ_conv = cconv(x_shifted, h_shifted,
   ↪ Nc);
46
47 % Cross correlation
48 cross_corr = xcorr(x_shifted, h_shifted);
49
50 % Auto correlation on x(-n+4)
51 auto_corr = xcorr(x_shifted);
52
53 % Plotting
54 figure;
55
56 subplot(3,2,1);
57 stem(n2, x_shifted, 'filled');
58 title('x(-n+4) [sin]');
59 xlabel('n'); ylabel('Amplitude');
60
61 subplot(3,2,2);
62 stem(n2, h_shifted, 'filled');
63 title('h(-n-3) [cos]');
64 xlabel('n'); ylabel('Amplitude');
65
66 subplot(3,2,3);
67 stem(0:length(lin_conv)-1, lin_conv,
   ↪ 'filled');
68 title('Linear Convolution');
69 xlabel('n'); ylabel('Amplitude');
70
71 subplot(3,2,4);
72 stem(0:length(circ_conv)-1, circ_conv,
   ↪ 'filled');
73 title('Circular Convolution');
74 xlabel('n'); ylabel('Amplitude');
75
76 subplot(3,2,5);
77 stem(-(length(x_shifted)-1):(length(h_shifted)-1),
   ↪ cross_corr, 'filled');
78 title('Cross Correlation');
79 xlabel('Lag'); ylabel('Amplitude');
80
81 subplot(3,2,6);
82 stem(-(length(x_shifted)-1):(length(x_shifted)-1),
   ↪ auto_corr, 'filled');
83 title('Auto Correlation of x(-n+4)');
84 xlabel('Lag'); ylabel('Amplitude');
85
86 sgtitle('Signal Operations: Transformed
   ↪ Sine and Cosine');

```

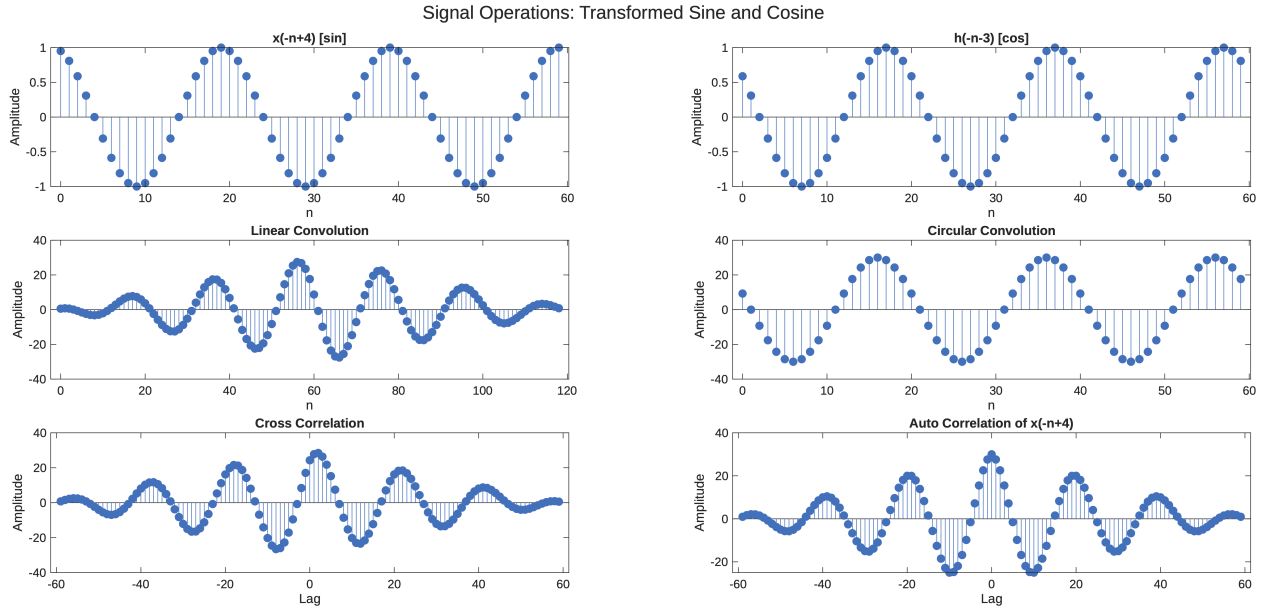


Figure 6: Operations on $x(-n+4)$ and $h(-n-3)$

Comments on Differences

- Linear: True result, longer output.
- Circular: Wraps, assumes periodicity.
- Cross-correlation: Shows similarity as signals shift.
- Auto-correlation: Peaks at zero lag, self-similarity.

Discussion

This experiment explored the practical differences between linear convolution, circular convolution, cross correlation, and auto correlation for discrete signals using MATLAB. By applying these operations to both arbitrary and sinusoidal signals, the distinct output characteristics were observed. Linear convolution produced longer output sequences, reflecting the full overlap between signals, while circular convolution maintained the original signal length due to periodic wrapping. Cross correlation revealed the degree of similarity and alignment between two signals, and auto correlation highlighted repeating patterns and periodicity within a single signal. The MATLAB results matched theoretical expectations, demonstrating how each operation serves specific purposes in signal analysis and processing.

Conclusion

The experiment successfully demonstrated the implementation and comparison of convolution and correlation operations on discrete signals using MATLAB. Linear and circular convolution, as well as cross and auto correlation, were applied to different signal types, and their outputs were analyzed. The results confirmed the theoretical properties of each operation and illustrated their relevance in digital signal processing tasks such as filtering, pattern recognition, and system analysis. This hands-on approach reinforced the importance of selecting the appropriate operation based on the application requirements.

References

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- [3] B. P. Lathi, R. Green, and Z. Ding, *Signal Processing and Linear Systems*, 2nd ed. Oxford University Press, 2017.